AI FOR ECONOMIC GROWTH & INDUSTRIAL PRODUCTIVITY

A Technical Presentation Delivered by

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PROTOCOLS

Your Excellency the Executive Governor of Katsina State, the Honourable Minister of Communications, Innovation and Digital Economy, the President, Nigeria Computer Society, distinguished Fellows and members of the NCS, honourable representatives of government and the security establishment, industry leaders, academics, investors, and all protocols duly observed — I am profoundly honoured to stand before this assembly.

DEFINITION

Economic growth, properly understood, is a sustained increase in real GDP per capita — the total value of goods and services produced by a nation, adjusted for inflation and divided by its population. But GDP is only a measure of market transaction value. It does not measure wellbeing, equity, or structural advancement. This distinction is not academic. Nigeria's GDP grew at an average of 7 percent annually between 2000 and

2014 — one of the fastest-growing economies in the world by that measure. And yet poverty rates barely moved. Infrastructure crumbled. Manufacturing shrank as a share of the economy. A nation can be simultaneously growing and declining. We have been that nation.

Economists have a more precise term for what actually matters: structural transformation — the process by which a society shifts from low-productivity agricultural activities to higher-productivity manufacturing and services. It is this structural shift that moved South Korea from a per capita income lower than Nigeria's in 1960 to one of the world's most advanced economies today. We have had growth. We have not had sufficient structural transformation.

Industrial productivity — the second concept in our title — refers to output per unit of input: how much a nation produces relative to the labour, capital, energy, and materials it uses. Economists measure it precisely as Total Factor Productivity (TFP) — the residual that cannot be explained by simply deploying more workers or more capital. It is the efficiency gain, the innovation premium, the intelligence multiplied into production. Studies of Nigerian manufacturing consistently show TFP growth that is near-zero or negative for extended periods. We have been working harder. We have not been working smarter. That is the precise problem that AI addresses.

DIAGNOSIS

Nigerians in the diaspora are among the highesteducated, highest-earning immigrant groups in the United States and

United Kingdom. Nigerian physicians, engineers, financiers, and academics routinely lead institutions in countries that were once our colonial masters. The same people. Different systems. The conclusion is inescapable: the problem was never primarily personal. It was structural.

Three structural forces have constrained Nigerian economic development:

The first is the resource curse — what economists call Dutch Disease. When Nigeria discovered oil in commercial quantities in 1956 and began producing at scale, the resulting export revenues caused the naira to appreciate significantly. This made everything Nigeria manufactured more expensive on global markets than imports. Factories shut down. By 2023, manufacturing contributed less than 5 percent of GDP — lower than the immediate post-independence era. We did not fail to industrialise because we lacked industrious people. We failed to industrialise because the very wealth we found underground made industry economically irrational.

The second is institutional failure. The economists Acemoglu and Robinson, in Why Nations Fail, distinguish between extractive institutions — designed to concentrate power and resources in few hands — and inclusive institutions — designed to distribute opportunity broadly. Their evidence across 500 years is unambiguous: extractive institutions produce economic stagnation regardless of natural resources or human talent. Nigeria's historical experience produced largely extractive institutions. Corruption and nepotism are not primarily moral failures of individuals. They are rational responses to extractive institutional environments where rule-following is not reliably rewarded.

The third is structural dependency. Prebisch and Singer demonstrated empirically that nations dependent on commodity exports face systematically deteriorating terms of trade over time — their exports buy progressively fewer manufactured goods. Nigeria has been a commodity exporter and has therefore faced precisely this dynamic. We did not choose this dependency. We inherited it from a colonial economic structure designed to make us raw material suppliers, not finished goods producers.

> *If the problem is structural, the solution must address structure — not merely uplift individual effort. And this is precisely where Artificial Intelligence enters with extraordinary relevance.*

One argument from the conventional account deserves preservation, because it is both true and theoretically grounded. Maslow demonstrated that human beings cannot reach self-actualisation — creative and innovative potential — when basic physiological and safety needs are unmet. When a Nigerian professional spends cognitive energy on generator fuel, borehole water, personal security, and healthcare gaps that the state should provide, that energy is not available for innovation. AI does not solve this problem entirely — but it can dramatically reduce the cognitive overhead of these burdens.

AGENERAL PURPOSE TECHNOLOGY

Artificial Intelligence is not merely a new product category. It is what economists call a General Purpose Technology — a class of innovations characterised by three properties: applicability across virtually every sector; continuous improvement over time; and the spawning of new industries beyond the original application. The canonical examples are the steam engine, electricity, and the internet. Each transformed not just its immediate application but the entire structure of economic possibility.

AI is the fourth General Purpose Technology of the modern era. Like its predecessors, it will not merely automate existing processes. It will transform the structure of what is economically possible. And like its predecessors, its full benefits arrive with a lag — requiring complementary investments in skills, institutions, and infrastructure. The nations that make those complementary investments early reap disproportionate rewards. Nigeria cannot wait.

Agentic AI — AI that takes autonomous sequences of actions — can substitute for the institutional reliability that our formal institutions often fail to provide. An algorithmically transparent loan assessment removes

the nepotism premium from credit markets. Recommendation systems solve information asymmetry problems: the farmer who does not know today's market price, the student who cannot access career guidance, the small manufacturer who cannot identify export markets. Time series forecasting enables the TFP gains our manufacturing and agriculture need: more output from the same inputs, through intelligence applied to production.

THE PRODUCER QUESTION

Ha-Joon Chang, in Kicking Away the Ladder, demonstrates that virtually every wealthy nation today achieved prosperity through deliberate industrial policy — directed investment in strategic industries, protection of infant industries, and technology acquisition from more advanced nations. Having climbed that ladder, they turned and told developing nations that free markets were the only legitimate path. The AI era reproduces this dynamic with precision.

Developing nations are being advised to simply use the AI tools that others' investments produce — to be consumers of AI, not producers. This is structural dependency in its most contemporary form: we supply the raw material (our data, our adoption, our attention) and others retain the valueadded activity (model training, algorithm design, platform ownership). The profit flows outward. Strategic control remains elsewhere.

Here is what Nigeria can and must produce:

Data as a sovereign asset — Africa holds linguistic data in 2,000+ languages barely represented in large models; agricultural data for crops unique to our climate; medical data for disease profiles shaped by our environment; economic behaviour data from some of the world's most dynamic informal markets. This data is currently being collected by foreign platforms at no cost to them. We must treat it as a national asset.

Models trained on African realities — a medical diagnosis model trained on North American data will underperform on Nigerian patients whose

disease profiles differ systematically. A credit assessment model trained on Western financial behaviour will mis-assess Nigerian informal sector workers whose creditworthiness is embedded in social networks.

Applications built for our markets — the highest-value AI is not the foundational model but the application layer built on top of it, tailored to specific contexts. And converted knowledge — our oral traditions, folk medicinal knowledge, agricultural practices refined over millennia — this is data of substantial economic value waiting to be structured.

> *The danger of remaining purely a consumer of AI built elsewhere has a name: techno-feudalism. Digital platforms reproduce feudal logic exactly. Build an audience on a platform you do not own and the platform owner can remove you. Store national data on foreign servers and those servers are governed by another country's laws. Own nothing and you are a tenant in someone else's digital economy.*

THE BACKBONE

An AI-driven economy runs on three non-negotiable physical foundations. Energy — reliable, affordable, high-volume electrical power. AI computation is energy-intensive. Nigeria's power generation challenges are a direct constraint on AI industrial capacity. Any serious national AI strategy must develop concurrently with a serious energy strategy. Storage and data sovereignty — Nigeria's data currently resides in data centres in Ireland, Virginia, Singapore, and Frankfurt, governed by other nations' laws. We must invest in domestic storage capacity and enforce data residency requirements for critical national data. Connectivity — Nigeria's rural connectivity remains deeply inadequate for the dataintensive applications that matter most: precision agriculture, remote healthcare, distributed learning, and security surveillance. Satellite communication is the fastest path to closing this gap.

AN AI SECURITY ARCHITECTURE

This summit was held here deliberately. Katsina has faced sustained attacks from bandits and insurgents that have killed thousands, displaced hundreds of thousands, and imposed a security tax on the entire economic life of the Northwest. The challenge is distinct: the population is not digitally active in ways that make conventional AI data collection possible. You cannot pull location data from people who do not use GPS. You cannot mine social media from communities not on social media.

The solution is to build the data architecture around how these communities actually live. Satellite imagery — free through Sentinel-2, commercial through Planet Labs, Nigerian through NASRDA — provides automated surveillance of high-risk zones without human presence. AI trained on baseline imagery detects newly cleared paths, fresh encampments, unusual vehicle accumulations. Mobile network metadata — not call content, but which tower, when, for how long — provides population-level movement mapping across the entire coverage area, legally accessible through security arrangements with telcos.

Market attendance patterns reveal community security perception with precision. In rural northern Nigeria, a half-empty market on a market day is a security signal, not an economic one. Inexpensive infrared footfall counters at market entrances generate attendance data that AI monitors against established baselines. Livestock GPS collars communicate what communities may not safely report: sudden scattering of herds, failure to return to established resting points, are distress signals from the land itself.

The Emirate structure — from ward head through district head to Emir — is the most powerful human intelligence network in the Northwest, almost entirely unmobilised for AI-integrated security purposes. A USSD-based reporting system requiring no smartphone, no internet connection, no literacy beyond basic numeracy, can convert this centuries-old human

network into a systematic, AI-processable intelligence feed. Community radio closes the loop: speech-to-text AI monitoring call-ins flags security-relevant reports in real time, and the same stations broadcast alerts outward.

> *We are not waiting for the population to become digital. We are building the data architecture around how they already live. All hardware — sensors, patrol drones, communication devices — must be designed and manufactured in Nigeria. This is what RAIN is committed to.*

FIVE NATIONAL MPERATIVES

Imperative One: Reorient AI Education from Use to Production

Teaching people to use AI applications produces consumers. Teaching people to build AI systems produces an industry. Our universities, polytechnics, and technical colleges must shift decisively toward the engineering of AI: data collection, model training, system integration, application development. AI-focused graduates should undergo a structured year of applied AI practice after their degrees — as law graduates attend law school, as medical graduates complete housemanship. RAIN has built this curriculum and is executing it. We ask government and the private sector to fund its national scaling.

Imperative Two: Direct Investment Toward Local AI Builders

Global AI is powered by private investment of extraordinary scale. Nigerian wealth holders have not invested proportionally in local AI infrastructure and companies. This summit must shift that perception. When we produce locally, we own the intellectual property. We pay salaries in Nigeria. We train Nigerian engineers. International partnerships must be evaluated on one criterion: does this transfer capacity and ownership, or create dependency?

Imperative Three: Establish Ethical and Regulatory Standards Now

AI deployed without ethical guardrails can entrench our institutional failures. A credit algorithm trained on biased historical data will reproduce that bias at scale. A predictive policing AI trained on historically unjust enforcement patterns will compound injustice. Nigeria's AI regulatory framework must be built before these systems are entrenched, not after. NITDA and relevant ministries must develop standards for transparency, bias auditing, data provenance, and accountability.

Imperative Four: Deploy AI Across the Productive Sectors

Education — AI assessment tools can mark essays, detect plagiarism, identify learning gaps, and generate personalised feedback for millions of students simultaneously. The most innovative young minds in Nigeria may be in classrooms too crowded for any teacher to notice them. Healthcare — AI diagnostic tools deployed at community health centres can bring specialist-level guidance to populations that currently have no specialists. Drug authentication AI can address the counterfeit medicine crisis. Agriculture — market-linkage platforms, crop disease detection, livestock geofencing, and yield forecasting can transform our most employment-intensive sector.

Imperative Five: Treat Infrastructure as Prerequisite, Not Afterthought

Every AI application in this speech depends on reliable energy, robust connectivity, and domestic data storage. A national AI strategy that does not begin with a serious energy strategy is a document written in sand. The government must treat AI infrastructure investment with the urgency it claims to apply to roads and bridges. The returns are higher.

CONCLUSION:

Nigeria's economic underperformance is not a story of insufficient effort or inadequate intelligence. It is a story of structural constraints that no amount of individual effort could overcome alone. Understanding this is not an excuse. It is the precondition for a correct solution.

Artificial Intelligence is a General Purpose Technology with the genuine potential to compensate for institutional weaknesses that have historically required decades to reform. An algorithm does not take a bribe. A satellite does not fear the forest. An AI-powered market platform does not require a broker with a cousin in the right place. These are genuine structural interventions — not rhetorical ones.

But AI will only serve this function if Nigeria engages with it as a producer — building its own models, on its own data, with its own engineers, governed by its own standards and its own strategic interests. The alternative is the reproduction of structural dependency in its most modern form. It is the digital equivalent of shipping crude oil and importing refined petroleum.

The window is open. The talent exists — our diaspora proves it. The data exists — our languages, our markets, our agriculture, our oral traditions, and yes, our security challenges are all data waiting to be structured and leveraged. The market exists — 220 million people with needs that foreign AI is poorly equipped to serve.

> *What is required is will: to invest before the returns are obvious, to build before the market demands it, to educate before the industry exists, and to regulate before the harms are entrenched.*

I thank you for your attention.

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